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Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

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Introduction

1.1 SymAware: High-level description of the project

SymAware addresses the fundamental need for a new conceptual framework for awareness in multi- agent systems (MASs) that is compatible with the internal models and specifications of robotic agents and that enables safe simultaneous operation of collaborating autonomous agents and humans. The goal of SymAware is to provide a comprehensive framework for situational awareness to support sustainable autonomy via agents that actively perceive risks and collaborate with other robots and humans to improve their awareness and understanding while fulfilling complex and dynamically changing tasks. The SymAware framework will use compositional logic, symbolic computations, formal reasoning, and uncertainty quantification to characterise and support situational awareness of MAS in its various dimensions, sustaining awareness by learning in social contexts, quantifying risks based on limited knowledge, and formulating risk-aware negotiation of task distributions. These objectives will be achieved in SymAware through

1. logical characterization of awareness using symbolic methods,
2. quantifying the symbolic reasoning for awareness with spatial and temporal ingredients for decision making,
3. risk awareness via quantified knowledge,
4. quantifying and communicating knowledge awareness,
5. demonstrating awareness engineering in aviation and automotive use cases, and
6. identifying requirements for ethical and trustworthy awareness in human-agent interaction.

The objectives of SymAware address the “Awareness Inside” Challenge of the European Innovation Council (EIC) by extending and formalizing human-based models of situational awareness and by providing a novel conceptual situational awareness framework for MASs that encompasses logical characterisation and integrative formal reasoning of interdependent awareness dimensions including knowledge, spatiotemporal, risk and social dimensions. This will support transitioning to safe mixed operation of autonomous agents and humans.

Second Ethical Report

To identify and mitigate any potential misuse of results and ethical issues, the SymAware project appointed an external ethical advisor, Dr. Matthijs Smakman, Associate Professor HU University of Applied Sciences Utrecht.

In collaboration with Dr. Smakman, a questionnaire was prepared at the time of the writing of the first ethical report (D9.4, June 2023), aimed at identifying the project partners that had conducted data collection in studies with human participants, and analyse any potential ethical issues related to the participation of human individuals and groups in experiments, the protection of personal data, and ethical issues linked to the deployment of the novel conceptual framework this project aims for.

This questionnaire was distributed in June 2023 and September 2023, and all six participating partners (MPI-SWS, TU/e, KTH, UU, NLR and SISW) completed the questionnaires. See Appendix A for the questionnaire template.

The results of the June questionnaire showed that only Uppsala University (UU) has worked with human individuals and has collected data from participants. Before their data collection involving human individuals they applied for ethical approval to the Swedish Ethical Review Authority (Etikprövningsmyndigheten); the application was evaluated as "not needing an ethical approval" due to the research not falling under the provisions of the Ethical Review Act of Sweden. No underaged or vulnerable participants were involved; consent forms were present and complete. See Appendix B for all answers of UU to the questionnaire related to the participation of human individuals and groups in experiments and the protection of personal data.

The results of the September questionnaire instead showed that no partners did any new data collection in studies with human participants in the period of 2023-06-01 until 2023-09-15.

A third questionnaire will be distributed to all partners again by the time of the Third Ethical Report.

APPENDIX A – TEMPLATE OF THE ETHICS QUESTIONNAIRE

For each of the three ethical reports that are to be prepared by the Ethical Advisor (Deliverables 9.4, 9.5, 9.6), each partner should fill out the standard form below.

The questions for this report (D9.X) refer to the time period between DD.MM.YYYY and DD.MM.YYYY.

If in this time period you didn't conduct any data collecting, answer all descriptive questions as "N/A".

Name and surname: _____

Email: _____

Partner Institution:

- **MPI-SWS**
- **TU/e**
- **KTH**
- **UU**
- **NLR**
- **SISW**

<u>Data acquisition</u>

Did you conduct activities for data acquisition (experiments, behavioural data, surveys, questionnaires, interviews, focus groups, etc)?

- **Yes**
- **No**

List all events in which you collected data. _____

For each of the events above, describe:
- Location and date of the event
- Indicate what personal data was collected/stored/processed:
- Data acquisition methods (cameras, microphones, behavioral data, physiological data, surveys, questionnaires, interviews, focus groups, mailing list, etc..).
- Anonymisation or pseudo--anomysation method (blurring), removal of data.
- Number of participants
- Methods of data processing (whether those operations are automated or not)
- Data to be kept (and for how long), data to be destroyed (if any) and how?
– Data and participant consent forms storage methods (saved on secured servers, protected by password, control access etc..)
- Strategy to handle subject access request
- Strategy to report data breaches

<u>Informed consent procedures and recruitment criteria</u>

Did you apply for ethical approval for research involving human participants or personal data? (if Yes, please elaborate) _____

Describe the criteria you used for identification of participants (in order to highlight any critical issues related to selection by gender, age, political opinion or religious, sexual orientation, etc). Describe inclusion and exclusion criteria. _____

Describe the procedures you used for the recruitment of participants. _____

Describe the procedures you used to inform participants about the purpose of the research. _____

In case of children/minors participants, describe the procedures you used to inform participants and their parents about the purpose of the research. _____

In case of children/minors participants, describe the procedures for obtaining approval from the guardian/legal representative and the agreement of the children or other minors. _____

In case of children/minors participants, what steps did you take to ensure that participants are not subjected to any form of coercion? A refusal must not have any consequence for the scholarship or other relationships with their teachers or other supervisory person.
What step did you take to ensure that the pupils who do not participate within a class (if they refuse or if their parents refuse to participate) will not feel discriminated? _____

Describe the procedures you used to inform participants about the use of collected information and the storage of data. _____

In case of children children/minors participants, describe the procedures you used to inform participants about the use of collected information and the storage of data. _____

Did you use a "informed consent document"? If yes, please email a copy of it to the Ethics Advisor Matthijs Smakman (matthijs.smakman@hu.nl). If not, why not? How did participants express their consent for the processing, collection and holding of personal data? _____

Describe the procedures you used to inform participants, including children, that they are free to withdraw from participation at any point. _____

In case your study had children/minors participants, please specify:
- Details of the age range.
- What were your assent procedures and parental consent for children and other minors?
- What steps did you take to ensure the welfare of the child or other minor?
- What justification was there for involving minors?

APPENDIX B - ANSWERS OF UU TO THE 1st QUESTIONNAIRE (June 2023)

List all events in which you collected data.

A participatory design study with users in the form of a workshop, 29/05-02/06.

- *Location and date of the event*
The study was conducted in Uppsala University, in the Social Robotics lab. There were four slots of 1hr30min, specifically 29/05 10:30-12:00, 30/05 10:30-12:00, 30/05 13:30-15:00, and 02/06 13:30-15:00.

- *Indicate what personal data was collected/stored/processed:*
We collected demographic data (age range, gender identity, driver experience), data related to the user's driving behaviour (separate questionnaire), and data related to the user's attitudes to autonomous vehicles (adapted from the NARS questionnaire on negative attitudes towards robots).
We also made audio recordings of the discussion between the participants and the experimenter.

- *Data acquisition methods (cameras, microphones, behavioral data, physiological data, surveys, questionnaires, interviews, focus groups, mailing list, etc..).*
Audio was recorded using an Uppsala University mobile phone. Questionnaire data was collected using three self-report questionnaires. The workshop data consisted of flip chart drawings and collage of post-its.

- *Anonymisation or pseudo--anomysation method (blurring), removal of data.*
The self-report questionnaires were filled using a randomized 8-character ID that participants generated themselves, their names and surnames are not saved with any of the collected data. Audio recordings are not anonymized though.

- *Number of participants*
In total there were 11 adult participants, split in 4 groups.

- *Methods of data processing (whether those operations are automated or not)*
Audio data will be manually coded and analyzed. Survey data will be manually analyzed in SPSS or another statistics software. Workshop data will be manually processed and analyzed.

- *Data to be kept (and for how long), data to be destroyed (if any) and how?*
Data will be stored on servers owned by Uppsala University. Collected data will be processed as long as necessary to ensure the quality of the research, and at least for the whole duration of the SymAware project.

- *Data and participant consent forms storage methods (saved on secured servers, protected by password, control access etc..)*
Data will be stored on servers owned by Uppsala University.

- *Strategy to handle subject access request*
Participants can request the audio recordings of their experimental slot. Survey data can be requested if participants retrieve it with their unique ID>

- *Strategy to report data breaches*
Incidents during the processing (data loses, intrusion, unwanted deletion etc.) will be reported to the data protection officer at Uppsala University (dataskyddsombud@uu.se).

Did you apply for ethical approval for research involving human participants or personal data?(if Yes, please elaborate)

Yes, we applied to the Swedish Ethical Review Authority (Etikprövningsmyndigheten). The application was evaluated as "not needing an ethical approval" due to the research not falling under the provisions of the Ethical Review Act of Sweden.

Describe the criteria you used for identification of participants (in order to highlight anycritical issues related to selection by gender, age, political opinion or religious, sexual orientation, etc). Describe inclusion and exclusion criteria.

Inclusion criteria was age of 18 years or above. No other criteria.

Describe the procedures you used for the recruitment of participants

Participants were recruited via advertisements on the Uppsala University notice boards. Additionally, advertisement was done via email in courses taught at the department of information technology. Participants then signed up via an online booking system (Doodle).

Describe the procedures you used to inform participants about the purpose of the research.

Participants were presented with an Information sheet that they could keep after the experiment. Any further questions were answered on the spot by the experimenter.

No minor participants.

Describe the procedures you used to inform participants about the use of collected information and the storage of data.

Participants were informed of this via the information sheet and the consent form.

Describe the procedures you used to inform participants, including children, that they are free to withdraw from participation at any point.

Participants were informed of this both verbally and via the information sheet.